Koneru Lakshmaiah Education Foundation
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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohd. Razia Alangir Banu
Student Name	Challagulla Shravya Chowdary
Roll Number	2520030322
Branch	CSE
Course Outcome	CO-2

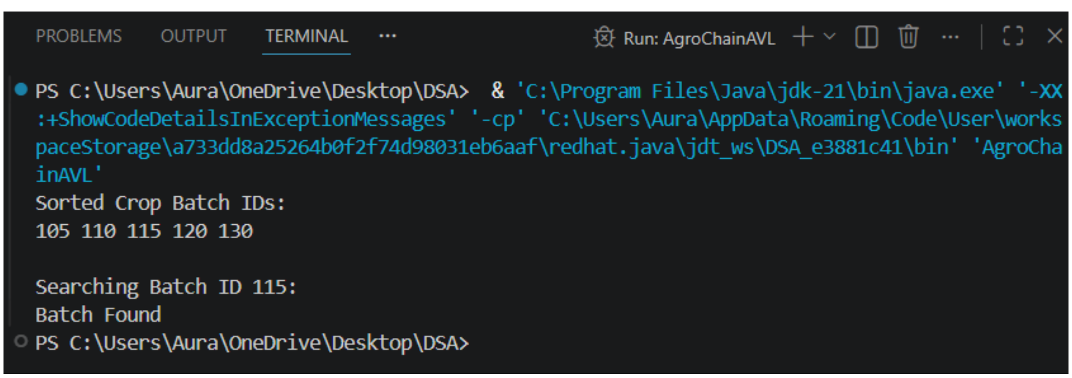
Case Study Topic: AgroChain – Smart Agriculture Supply Chain Management System

Work Done Summary:
Developed the AgroChain Agricultural Supply Record Management module using **B-Trees, B+ Trees, Segment Trees, and Fenwick Trees**. Agricultural supply records were indexed efficiently to support fast searching and retrieval. B+ Trees were utilized for range-based queries on crop prices and harvest dates, while Fenwick Trees were implemented to calculate cumulative crop production statistics. The system improved data organization, analytical processing, and overall supply chain efficiency.

Implementation Details:

- Used **B-Tree** for storing and indexing agricultural supply records.
- Implemented **B+ Tree** for efficient range queries on crop prices and harvest dates.
- Used **Fenwick Tree (Binary Indexed Tree)** to calculate cumulative crop production.
- Applied range query operations to analyse agricultural data.
- Improved search efficiency and scalability for large datasets.

Result: screenshot/output:



- Agricultural data was stored and processed efficiently.
- Cumulative production statistics were calculated accurately.
- Range queries reduced processing time.
- The system supported fast analytics and decision-making.
- AgroChain achieved efficient management of large agricultural datasets.

GitHub Link: **https://github.com/challagullashravya-tech/DSA-2**

NOTE: The Code and Screenshots have been uploaded in the above link.

Student Signature	Faculty Signature
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